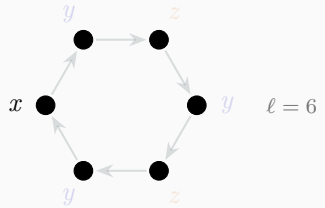
An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

Yuya Uezato Yoshiki Nakamura
CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

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FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



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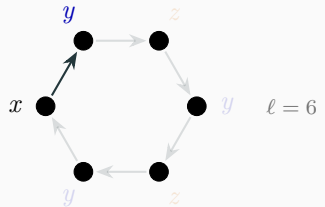
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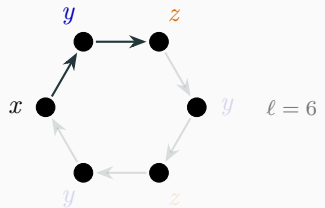
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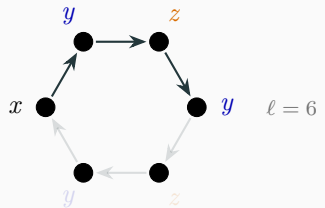
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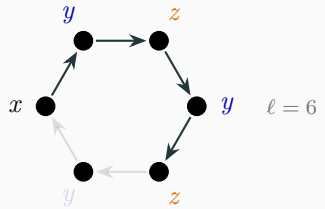
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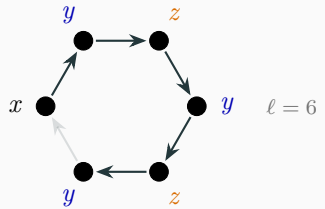
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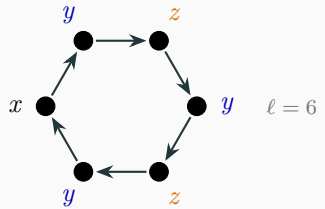
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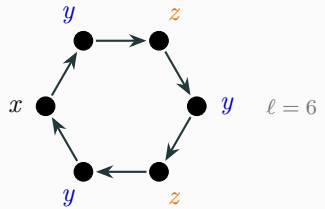
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We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

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To obtain the form of $\exists z.\ \cdots$, we use DNF translation.
$\rightsquigarrow 2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22, IEAMF].

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Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

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(This translation uses 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 *to* CoR*?*

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

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- 例：k 個の論理和の論理積を DNF にすると、合成が 2^k 個
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Outline

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Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

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- 答えは YES：$|\varphi_k| = O(k)$ で、同値な CoR 項は合成 $\geq 2^{k-1}$ 個
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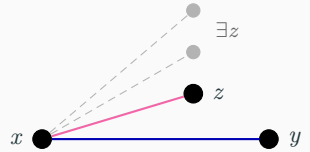
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The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



$G_k(x,y)$ holds iff there is z such that
for every $i \in [k]$,
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have labels P_i and Q_i, in either order.

CoR: 2^k compositions

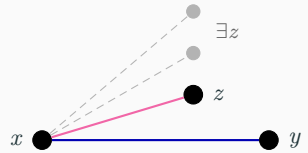
$$G_k \equiv \bigcup_{A \subseteq [k]} \left[\left(\bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left(\left(\bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$

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- G_k：z を選んで、各 i で x-y 辺と x-z 辺の一方が P_i、他方が Q_i
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Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
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☞ If $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

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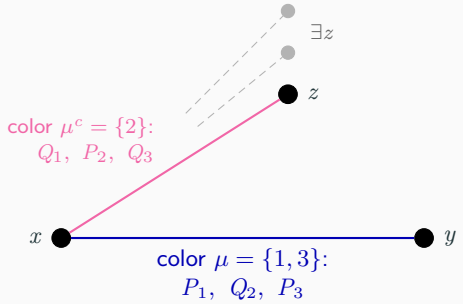
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Structures as colored complete graphs

View structures as **complete graphs**, colored with 2^k colors $\mu \subseteq [k]$: $P_i \iff i \in \mu$.

Q_i is used as the complement of P_i (i.e., $i \notin \mu$): exactly one of P_i, Q_i holds on each edge.



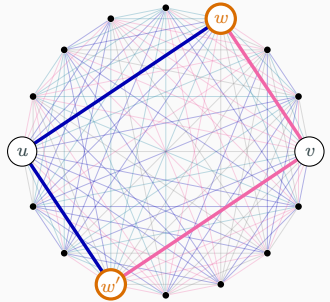
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have labels P_i and Q_i, in either order

👉 $G_k(x, y)$ holds iff some x-z edge has the color μ^c, where μ is the color of the x-y edge.

ページ概要：

- 構造を 2^k 色の色付き完全グラフとして見る（色 $\mu \subseteq [k]$、$P_i \iff i \in \mu$）
- $G_k(x, y) \iff$ x-y 辺の色 μ の補 μ^c の色の x-z 辺がある

The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k4^k)$ vertices,
cf. [JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** w such that the u-w and w-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

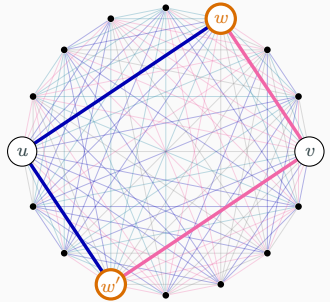
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The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k4^k)$ vertices,
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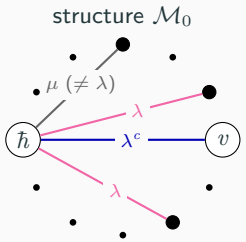
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$\hbar$ has λ-edges

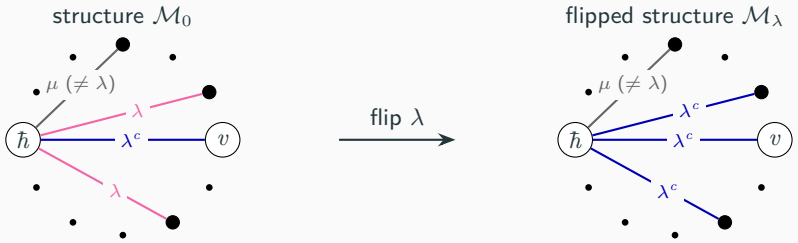
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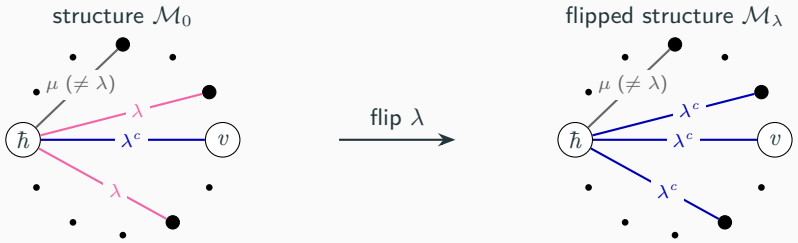
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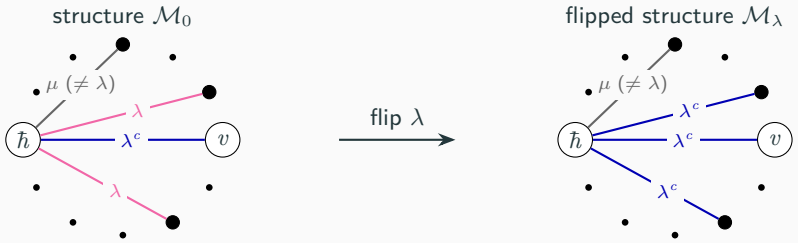
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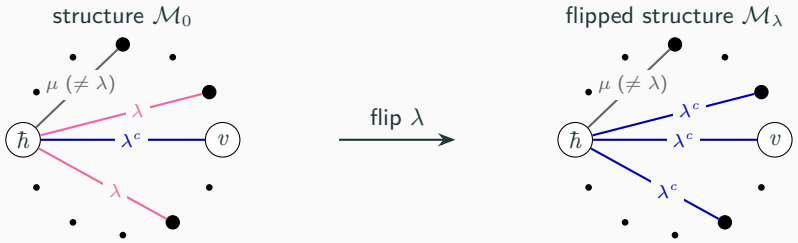
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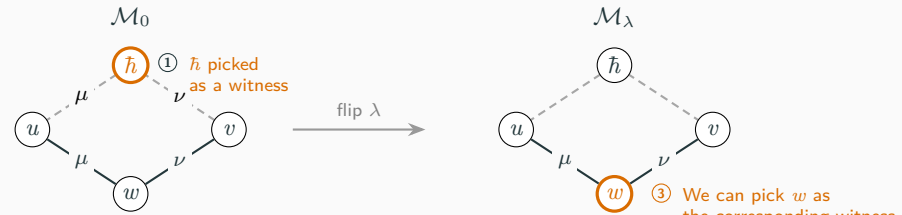
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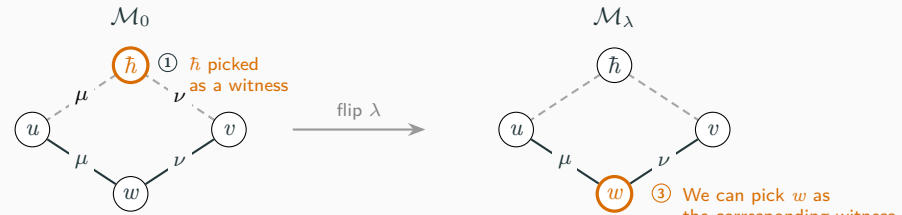
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Case $u = \hbar$ (similarly for $v = \hbar$): the $\hbar$-w edge itself may be flipped.

So the answer changes only if t holds on some λ-edges and only on them ($v = \hbar$: the same for s).
Such λ is uniquely determined for t and for s.
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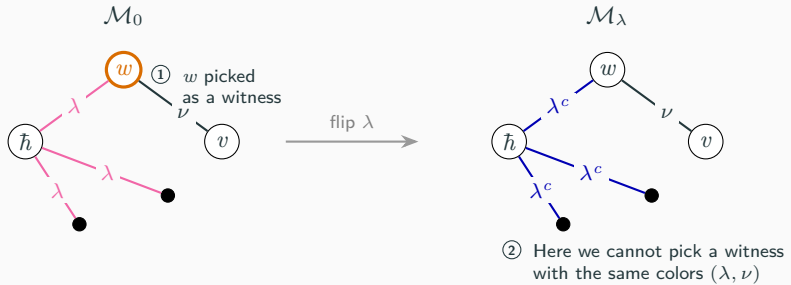
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- （補足）$(\hbar, \hbar)$ や端点の witness（$w \in \{u, v\}$）も同じ考え方で処理できる（論文参照）

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (2/2)

Case $u = \hbar$ (similarly for $v = \hbar$): the $\hbar$-w edge itself may be flipped.



So the answer changes only if t holds on some λ-edges and only on them ($v = \hbar$: the same for s).
Such λ is uniquely determined for t and for s.
$\rightsquigarrow$ Each composition newly distinguishes at most two $\mathcal{M}_\lambda$.
(The remaking cases: the pair $(\hbar, \hbar)$ and witnesses $w \in \{u, v\}$. See the paper for details.)

ページ概要：

- $u = \hbar$：hub の λ-edges は全て λ^c になる → 色の組 (λ, ν) の witness は $\mathcal{M}_\lambda$ で取れない
- t が λ 以外の色でも成り立てば別の witness が取れる。答えが変わるのは t が λ-edges だけで成り立つときだけ（$v = \hbar$ は s について同様）
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Outline

1. Our results

2. Technical overview

3. Conclusion

Conclusion

Thm. (This paper)

There are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Our proof also applies to FO^3 vs. FO^2, matrix query languages, and CoR with RTC.

Key Idea the randomly colored structure $\mathcal{M}_0$ with its 2^k flipped structures.
Each composition can newly distinguish ≤ 2 of them.

Problem (Open Questions)

- *a superpolynomial gap over a* ***fixed*** *finite signature (e.g., one binary relation)?*
- *the domain size from which the blowup begins?*
- *the exact number of compositions for* G_k*: between* 2^{k-1} *and* 2^k*?*

Thank you for your attention!

ページ概要：

- 結果：$|\varphi_k| = O(k)$ の FO^3 式に同値な CoR 項は合成 $\geq 2^{k-1}$ 個。同じ手法が FO^2、行列言語、RTC 付き CoR にも効く
- 鍵：ランダム彩色 $\mathcal{M}_0$ の 2^k 通りの flip。各合成は高々 2 個しか新たに見分けない
- 未解決：固定シグネチャでの差、爆発が始まる領域サイズ、G_k の正確な合成数

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References i

[Var95] M. Y. Vardi. "On the Complexity of Bounded-Variable Queries". In: **Proceedings of the Fourteenth ACM SIGACT-SIGMOD-SIGART Symposium on Principles of Database Systems (PODS 1995)**. ACM, 1995, pp. 266–276.

[BS21] K. Bringmann and J. Slusallek. "Current Algorithms for Detecting Subgraphs of Bounded Treewidth Are Probably Optimal". In: **48th International Colloquium on Automata, Languages, and Programming (ICALP)**. Vol. 198. Leibniz International Proceedings in Informatics (LIPIcs). Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2021, 40:1–40:16. DOI: 10.4230/LIPIcs.ICALP.2021.40.

[NU27] Y. Nakamura and Y. Uezato. "A Fine-Grained Dichotomy for Bounded-Variable Query Evaluation: The Calculus of Relations, a Boolean Modal Logic, and One-Variable Counting Logic". In: **Proceedings of the ACM on Management of Data** 4.5 (PODS) (2027), 295:1–295:27. DOI: 10.1145/3850130.

[Wil14] R. Williams. "Faster Decision of First-Order Graph Properties". In: **Joint Meeting of the Twenty-Third EACSL Annual Conference on Computer Science Logic and the Twenty-Ninth Annual ACM/IEEE Symposium on Logic in Computer Science (CSL-LICS)**. ACM, 2014, 80:1–80:6. DOI: 10.1145/2603088.2603121.

References ii

[Tar41] A. Tarski. "On the Calculus of Relations". In: **The Journal of Symbolic Logic** 6.3 (1941), pp. 73–89. DOI: 10.2307/2268577.

[TG87] A. Tarski and S. Givant. **A Formalization of Set Theory without Variables**. Vol. 41. Colloquium Publications. Providence, RI: American Mathematical Society, 1987. DOI: 10.1090/coll/041.

[Nak22] Y. Nakamura. "Expressive Power and Succinctness of the Positive Calculus of Binary Relations". In: **Journal of Logical and Algebraic Methods in Programming** 127 (2022), p. 100760. DOI: 10.1016/j.jlamp.2022.100760.

[FGL+15] G. H. L. Fletcher, M. Gyssens, D. Leinders, D. Surinx, J. Van den Bussche, D. Van Gucht, S. Vansummeren, and Y. Wu. "Relative Expressive Power of Navigational Querying on Graphs". In: **Information Sciences** 298 (2015), pp. 390–406. DOI: 10.1016/j.ins.2014.11.031.

[Nak23] Y. Nakamura. "Existential Calculi of Relations with Transitive Closure: Complexity and Edge Saturations". In: **38th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)**. IEEE, 2023, pp. 1–13. DOI: 10.1109/LICS56636.2023.10175811.

References iii

[VS24] H. Vinall-Smeeth. "From Quantifier Depth to Quantifier Number: Separating Structures with k Variables". In: **Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)**. Full version: arXiv:2311.15885. ACM, 2024, 71:1–71:14.

[GS05] M. Grohe and N. Schweikardt. "The Succinctness of First-Order Logic on Linear Orders". In: **Logical Methods in Computer Science** 1.1:6 (2005). Conference version in LICS 2004, pp. 1–25. DOI: 10.2168/LMCS-1(1:6)2005.

[BGV19] R. Brijder, M. Gyssens, and J. Van den Bussche. **On Matrices and K-Relations**. arXiv:1904.03934v1. 2019. arXiv: 1904.03934 [cs.DB]. URL: https://arxiv.org/abs/1904.03934v1.

[BGV22] R. Brijder, M. Gyssens, and J. Van den Bussche. "On Matrices and K-Relations". In: **Annals of Mathematics and Artificial Intelligence** 90.2–3 (2022), pp. 181–210. DOI: 10.1007/s10472-021-09760-4.

[JMT95] P. Jipsen, R. D. Maddux, and Z. Tuza. "Small Representations of the Relation Algebra $E_{n+1}(1, 2, 3)$". In: **Algebra Universalis** 33 (1995), pp. 136–139. DOI: 10.1007/BF01190770.